HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered **Python** scripts to automate data ingestion and ETL processes for financial reporting, reducing manual effort by an estimated 15 hours per week.
- Developed and maintained **Java Spring Boot** REST APIs for internal services, handling approximately 500 requests per minute with a p95 latency under 200ms.
- Managed production incidents for enterprise applications, performing root cause analysis and resolving P1/P2 issues within defined SLAs, contributing to a 98% system availability.
- Designed and simulated **DSP** and **FEC algorithms** in **MATLAB** and **Simulink** to optimize free space optical communication link performance and minimize power consumption.
- Collaborated with the **FPGA** engineering team to define fixed-point models and specifications for translating complex DSP algorithms into RTL implementations.
- Developed end-to-end system models of optical components including **lasers**, **modulators**, and **photodetectors** in **MATLAB** to predict system performance under various environmental conditions.
- Partnered with optical testing and firmware teams to implement automated algorithms for transceiver calibration and runtime optimization in dynamic space environments.

SKILLS

DSP & Communications – Digital Signal Processing, Forward Error Correction, Link Performance Optimization, Communication Theory, Modulation, Noise Modeling, Fixed-Point Arithmetic

Simulation & Modeling – MATLAB, Simulink, System Modeling, Algorithm Development, Floating-Point Precision, Fixed-Point Precision

Programming & Scripting – Python, C/C++, Java, Spring Boot, REST APIs

Hardware Interaction – FPGA, RTL Implementation, Hardware Co-Design, Optical Systems, Lasers, Modulators, Photodetectors

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Jenkins, Git

Data & Analytics – SQL, PostgreSQL, MySQL, Oracle, pandas, NumPy, scikit-learn, Power BI, DAX

Operations & Support – Production Support, Incident Response, SLA Management, Splunk, Dynatrace, Linux, Shell Scripting

PROJECTS

DSP Algorithm Optimization for Optical Links

- Designed and simulated advanced **DSP algorithms** in **MATLAB** to enhance signal-to-noise ratio in free space optical communication, achieving a simulated 15% improvement in link budget.
- Developed **fixed-point models** of key DSP components using **Simulink** to assess resource requirements and feasibility for FPGA implementation, ensuring bit-true accuracy.
- Collaborated with a hardware team to translate floating-point algorithm specifications into detailed requirements for RTL development, facilitating a smoother integration process.

Automated Optical Transceiver Calibration System

- Built **Python** scripts to automate the calibration process for optical transceivers, reducing manual calibration time by approximately 40% for each unit.
- Integrated test equipment control software with the calibration scripts to manage laser power, modulator bias, and photodetector gain settings.
- Developed data logging and reporting mechanisms to track calibration results and identify trends for potential hardware improvements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)